MANENDRA SINGH

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong problem-solving skills and hands-on experience in software development, data structures & algorithms, machine learning, and full-stack application development. Solved 400+ LeetCode and 300+ GeeksforGeeks problems, demonstrating strong analytical and coding abilities. Experienced in Java, Python, SQL, JavaScript, Node.js, and MySQL, with use of AI tools to accelerate development, debugging, and deployment while maintaining a test-driven, quality-first mindset. Passionate about building scalable, real-world solutions, learning emerging technologies, and contributing effectively in collaborative, fast-paced environments.

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu 2023 – 2027
Bachelor of Technology in Computer Science and Engineering CGPA: 8.52

Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan 2021 – 2022
Higher Secondary School (12th Standard) Percentage: 91.4

Swami Keshwanand Shikshan Sansthan, Sikar, Rajasthan 2019 – 2020
Secondary School (10th Standard) Percentage: 88.33

PERSONAL PROJECTS

AI-Powered Teledermatology Platform for Acne Severity Detection GitHub

- An AI-powered web app that analyzes patient photos to instantly determine the severity of their acne and automatically alert doctors for high-risk cases
- Tech Stack: Python, Flask, TensorFlow, AWS (Amplify, EC2, S3, DynamoDB, SNS, Cognito), Cropper.js
- Engineered a deep learning inference engine using transfer learning (EfficientNet) to classify 5 acne subtypes, paired with a Cropper.js frontend to isolate regions of interest and eliminate noise.
- Architected a stateless microservices backend deployed on AWS EC2, utilizing Amazon S3 for decoupled image storage and DynamoDB for logging diagnostic clinical records.
- Implemented secure patient authentication via AWS Cognito and an event-driven triage system using Amazon SNS to instantly alert dermatologists of severe, high-risk conditions.

Diabetic Retinopathy Detection (Live Demo) GitHub

- A clinical AI web app for automated detection and classification of diabetic retinopathy stages from retinal fundus images.
- Tech Stack: Python, TensorFlow, EfficientNet, ResNet, OpenCV, Grad-CAM, Streamlit, FPDF
- Trained deep learning models (EfficientNet, ResNet, custom CNN) using transfer learning with data augmentation and class imbalance handling to classify DR severity into 5 stages.
- Integrated Grad-CAM heatmap visualization to highlight pathological retinal regions, improving model interpretability.

SCSMS – Smart City Service Management System (Live Demo) GitHub

- A real-time smart city platform for streamlined service reporting and citizen-admin communication.
- Tech Stack: Node.js, Express, MySQL (Aiven), Socket.IO, Leaflet.js, OpenStreetMap API, JavaScript
- Built a scalable and secure backend system enabling real-time communication, reducing issue reporting latency by 40% using Socket.IO.
- Designed a secure relational database schema with role-based access control (RBAC), indexing, and normalized table structures to ensure data integrity and efficient query resolution.
- Developed RESTful APIs following MVC architecture, ensuring maintainability, modularity, and adherence to software engineering best practices.

TECHNICAL SKILLS

- **Programming Languages:** Java, C, C++, Python, SQL, JavaScript
- **Web Tech:** Node.js, Express.js, HTML, CSS
- **Databases & Cloud:** MongoDB, MySQL, Firebase (Cloud-based DB)
- **Core CS Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems
- **Software Engineering:** REST APIs, MVC Architecture, Unit Testing, Scalable Systems, Secure Coding, Debugging
- **Developer Tools:** Git, GitHub, Visual Studio Code

CERTIFICATIONS & ACHIEVEMENTS

- **LeetCode:** Resolved 450+ problems — Peak Rating: 1783 — Global Rank $<$ 2,50,000 (Profile)
- **GeeksforGeeks:** Resolved 300+ problems — Global Rank $<$ 40,000 — Institute Rank $<$ 500 (Profile)
- **Generative AI - IBM** (Proof)
- **Machine Learning & Data Science** – Skill Up, GeeksforGeeks / Nation SkillUp (Proof)
- **AI Agent Workshop using AgentForce** – GeeksforGeeks & Salesforce (July 2025) (Proof)
- **Google Cloud Agentic AI Day** – Hack2Skill (July 2025) (Proof)
- **Adobe India Hackathon** – Adobe (July 2025) (Proof)